

Combined Families Summary

LLM · ASR · TTS bake-off — cross-family overview

Prepared 23 July 2026 · Egyptian Arabic voice robot stack

LLM (speed)

Nile-Chat-4B

LLM (quality)

Qwen3-8B

ASR pick

Whisper-Large-v3-Turbo-CT2

TTS pick

VoiceTut-TTS

Executive verdict

Recommended package: Whisper-Large-v3-Turbo-CT2 + Nile-Chat-4B + VoiceTut-TTS (~13.7 GB peak VRAM sum). Quality package: Whisper-Large-v3-Turbo-CT2 + Qwen3-8B + VoiceTut-TTS (~12.2 GB peak VRAM sum).

Overview

This summary helps select one model from each family for the Arabic voice robot stack.

All three families now have successful bake-off results and shortlists.

Suggested ASR default: Whisper-Large-v3-Turbo-CT2. Accuracy alternative: Whisper-Large-v3-CT2.

Suggested LLM speed default: Nile-Chat-4B. Quality alternative: Qwen3-8B.

Suggested TTS default (speed/resources): VoiceTut-TTS - confirm by listening to the WAV.

Prefer LOWER latency/memory/error rates; prefer HIGHER accuracy/throughput/quality/robot scores.

How to read this report

Page 2: shared metrics glossary.

Page 3: shortlist table and decision path.

Page 4: combined GPU / VRAM stack budget (critical).

Page 5: family readiness and top scores.

Pages 6-8: LLM, ASR, and TTS snapshot charts.

Page 9: package contents and bottom line.

Combined - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

TTFT - Time To First Token (s) LOWER better Delay until the model starts generating. Lower = snappier turn-taking (robot starts sooner).	Peak VRAM (MB) LOWER better GPU memory used at peak. Lower leaves room to run ASR + LLM + TTS together on one GPU.
Throughput (tokens/s) HIGHER better How fast tokens are generated after the first one. Higher = long answers finish sooner.	Peak RAM (MB) LOWER better System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.
Generate time (s) LOWER better Total time to finish the reply. Lower is better (also depends on answer length).	Peak CPU % LOWER better CPU utilization spike during the run. Lower means less contention with other services.
Accuracy % (word accuracy) HIGHER better Share of words recognized correctly (about 1 - WER). Higher = fewer recognition mistakes.	Cold load time (s) LOWER better Seconds to load the model into memory the first time. Matters for startup / cold starts.
WER - Word Error Rate LOWER better Fraction of words wrong. 0 = perfect. Lower = better recognition.	RTF (Real-Time Factor) LOWER better Processing time / audio duration. RTF < 1 = faster than realtime (good). RTF > 1 = slower than the audio.
CER - Character Error Rate LOWER better Same idea as WER at character level. Useful for Arabic. Lower = better.	x Realtime (xRT) HIGHER better How many times faster than realtime (about 1/RTF). Example: 6x = 6s of audio in ~1s of compute.
Chars/s (characters per second) HIGHER better Input characters synthesized per second of compute. Higher = faster speech generation.	Robot realtime score (0-100) HIGHER better Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Remember

Shared rule: LOWER = TTFT, RTF, WER, CER, VRAM, RAM, CPU%, load/generate/transcribe time.

Shared rule: HIGHER = accuracy%, tokens/s, chars/s, quality scores, robot/balanced scores, xRealtime.

See each family PDF glossary for the full term list.

Recommended shortlist

ASR — robot default

Whisper-Large-v3-Turbo-CT2

Best accuracy/speed/VRAM blend for realtime input

ASR — accuracy alt

Whisper-Large-v3-CT2

Highest word accuracy when recognition quality is critical

LLM — speed default

Nile-Chat-4B

Best TTFT/throughput/VRAM composite for turn-taking

LLM — quality alt

Qwen3-8B

Best rubric quality (4.65/5) and auto-pass rate

TTS — speed/resources

VoiceTut-TTS

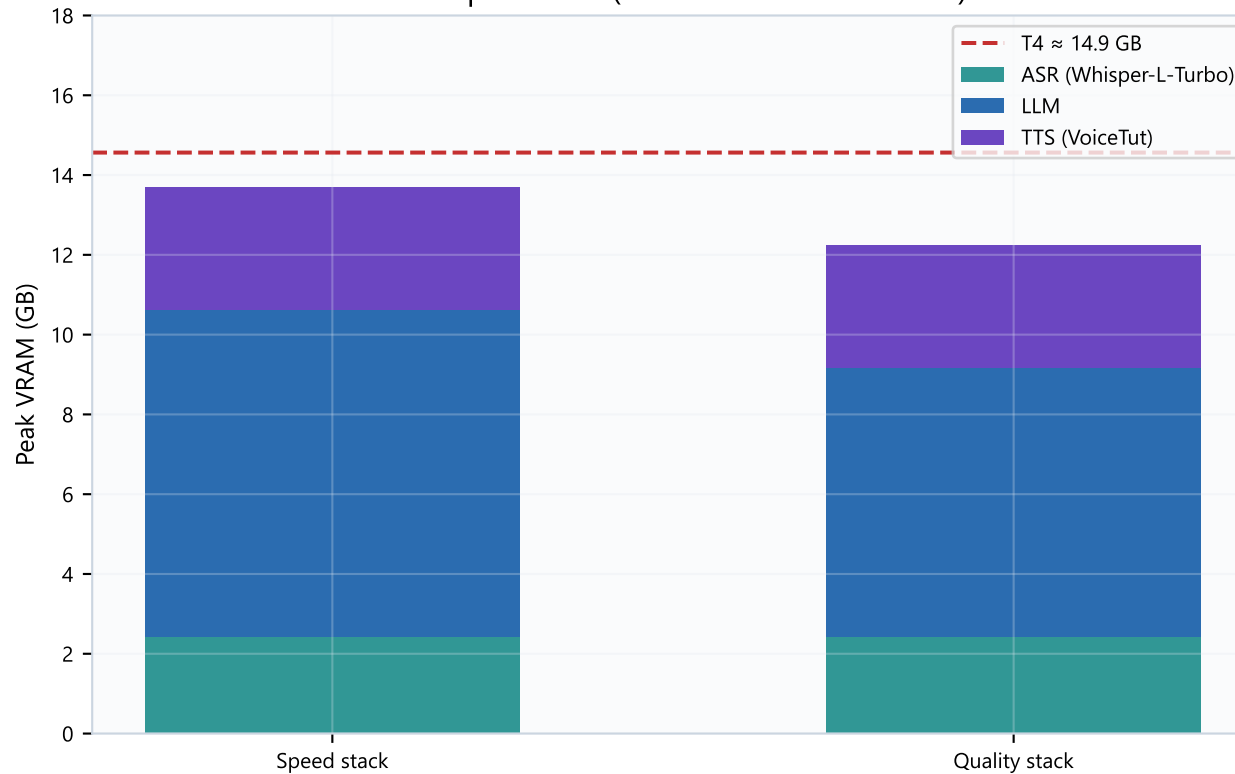
Best RTF/throughput/VRAM composite — listen to WAV before freeze

Decision path

- 1) Lock ASR first — input quality gates the whole pipeline.
- 2) Lock LLM next — choose speed (Nile-Chat-4B) or quality (Qwen3-8B / Qwen3-4B).
- 3) Lock TTS after listening — default VoiceTut-TTS; fallback SILMA-TTS if voice quality wins.
- 4) Validate the combined VRAM budget on the deployment GPU (next page).

Combined GPU stack budget (peak VRAM)

Naive peak sum (worst case if all resident)



Budget numbers

ASR: 2.43 GB
 LLM speed: 8.20 GB
 LLM quality: 6.74 GB
 TTS: 3.08 GB

Speed sum: 13.71 GB
 Quality sum: 12.25 GB
 T4 usable: 14.6 GB

Headroom is tight on speed stack.
 Prefer quality LLM (int4) if co-resident.

How to read the stack budget

Speed package = Whisper-Large-v3-Turbo-CT2 + Nile-Chat-4B + VoiceTut-TTS $\approx$ 13.7 GB peak sum.

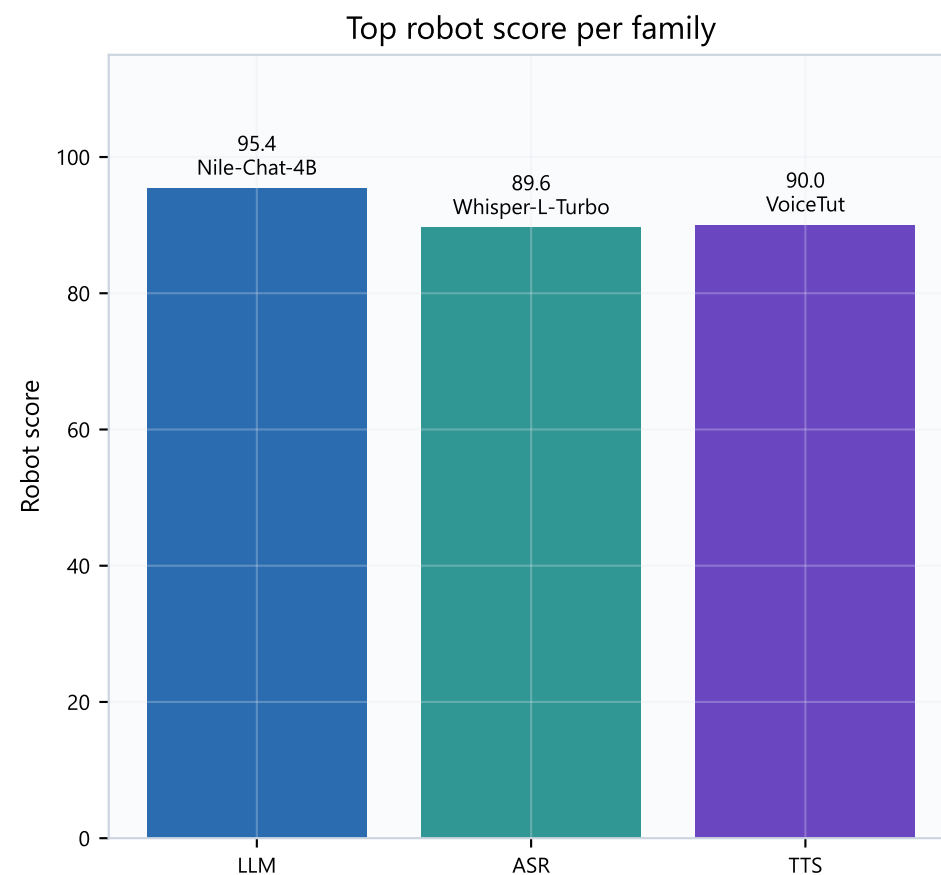
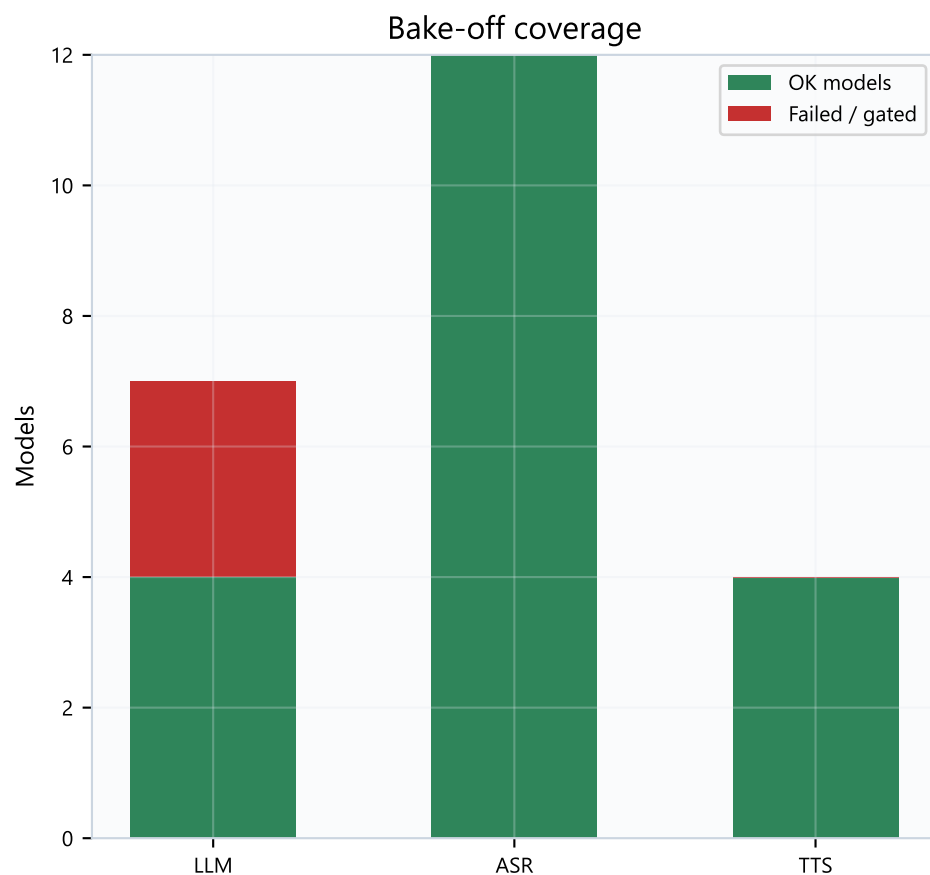
Quality package = Whisper-Large-v3-Turbo-CT2 + Qwen3-8B + VoiceTut-TTS $\approx$ 12.2 GB peak sum.

These are peak-per-family sums — true concurrent residency may be lower if models swap, but plan for the worst case.

On T4-class GPUs, prefer the quality LLM (Qwen3-8B int4) or Whisper-Small if headroom is insufficient.

Always measure actual concurrent VRAM on the deployment box before freeze.

Family readiness & top robot scores



Readiness reading

ASR is fully covered (12/12 OK). LLM has 4 OK models + 3 gated failures.

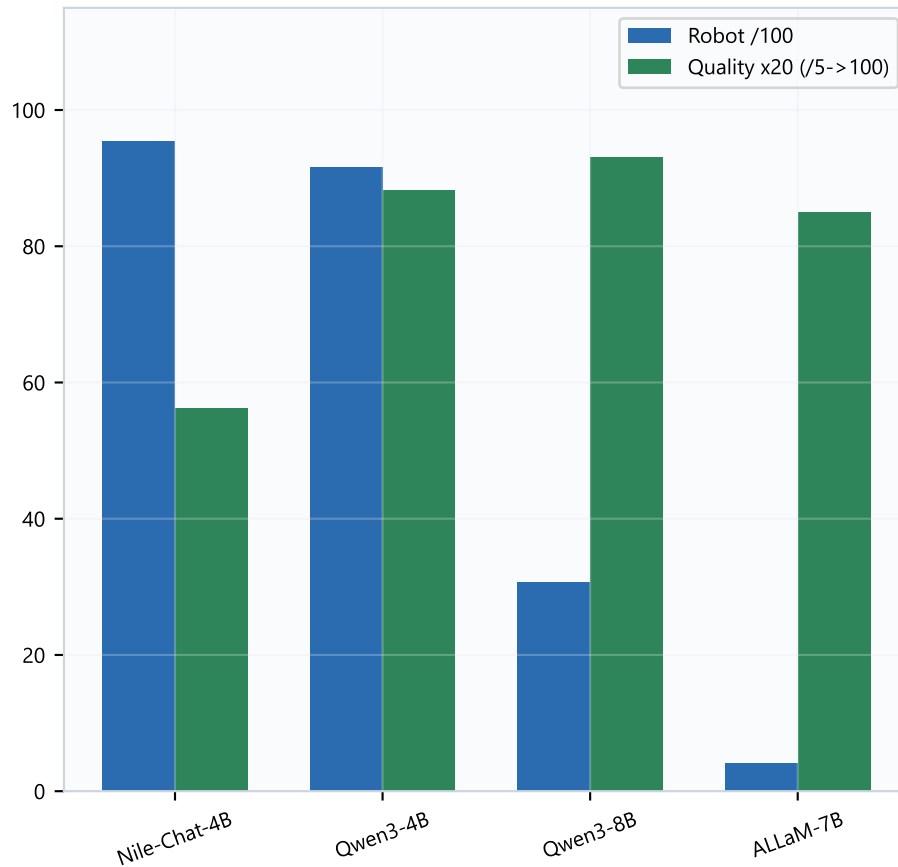
TTS is ready for shortlisting (4/4 OK) — VoiceTut leads robot score.

All three families can be shortlisted together; only TTS still needs a human listening pass.

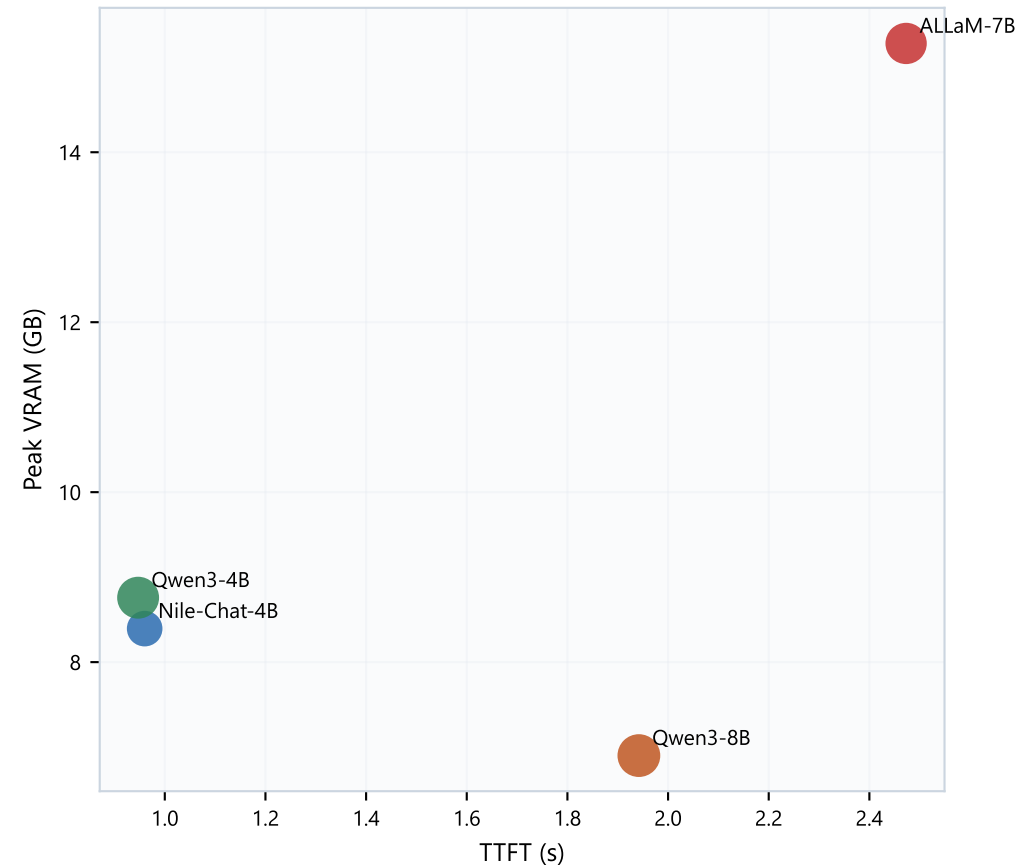
Plan GPU headroom for Whisper + LLM + VoiceTut/SILMA together (see stack budget page).

LLM snapshot

Speed composite vs quality



Bubble size ~ quality

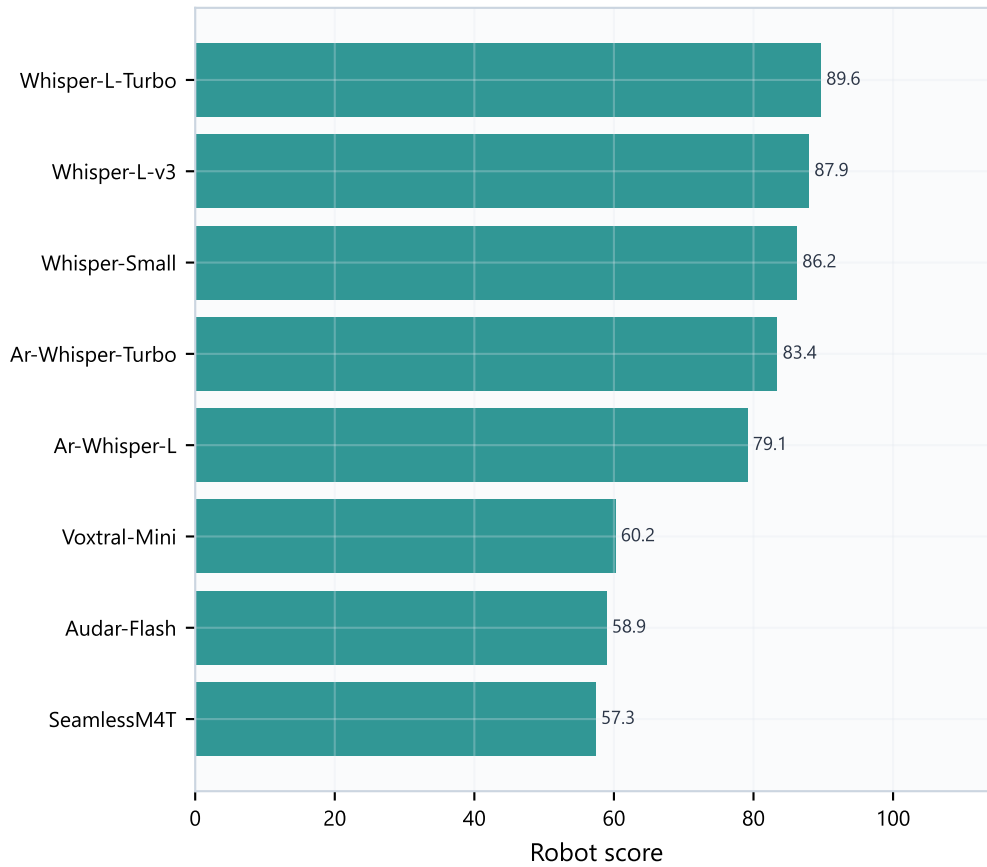


LLM takeaway

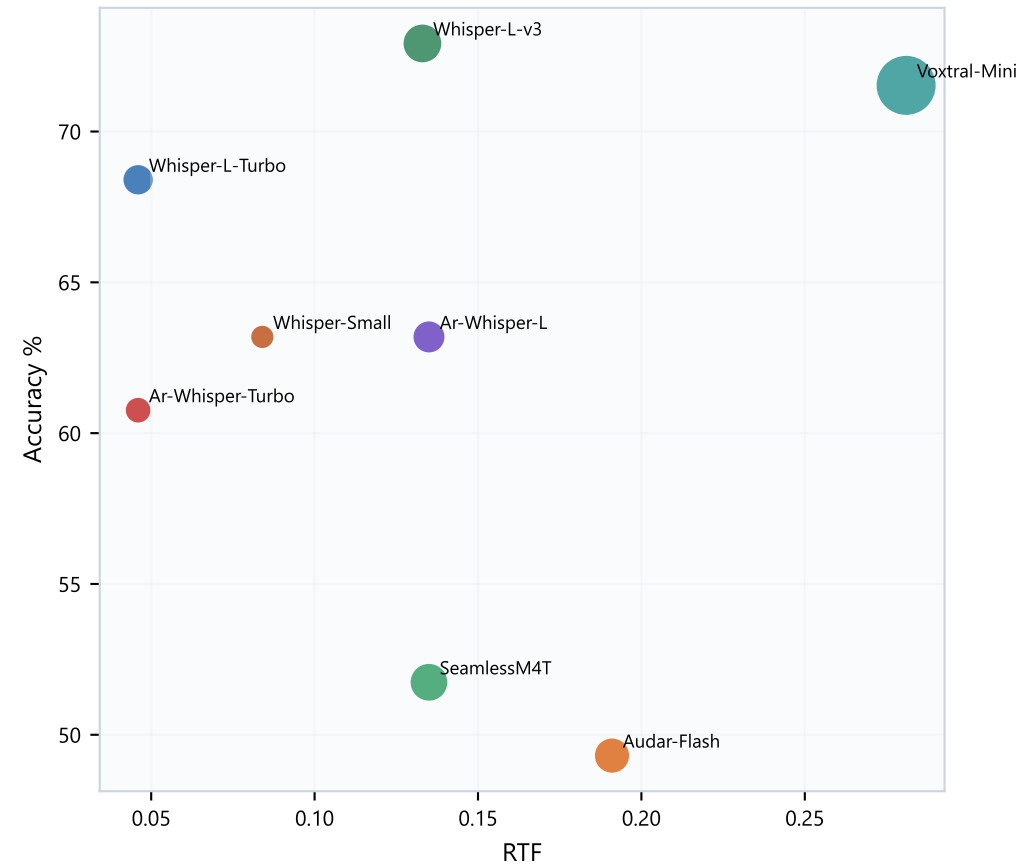
Nile-Chat-4B wins robot realtime; Qwen3-8B wins quality.
If spoken answers must stay short and correct, lean quality (Qwen).
If the robot must start speaking immediately, lean Nile-Chat-4B — then spot-check verbosity.
Qwen3-4B is the practical middle ground for many deployments.

ASR snapshot (top 8)

Robot leaderboard



Accuracy vs speed (bubble = VRAM)



ASR takeaway

Default: Whisper-Large-v3-Turbo-CT2 for robot balance.

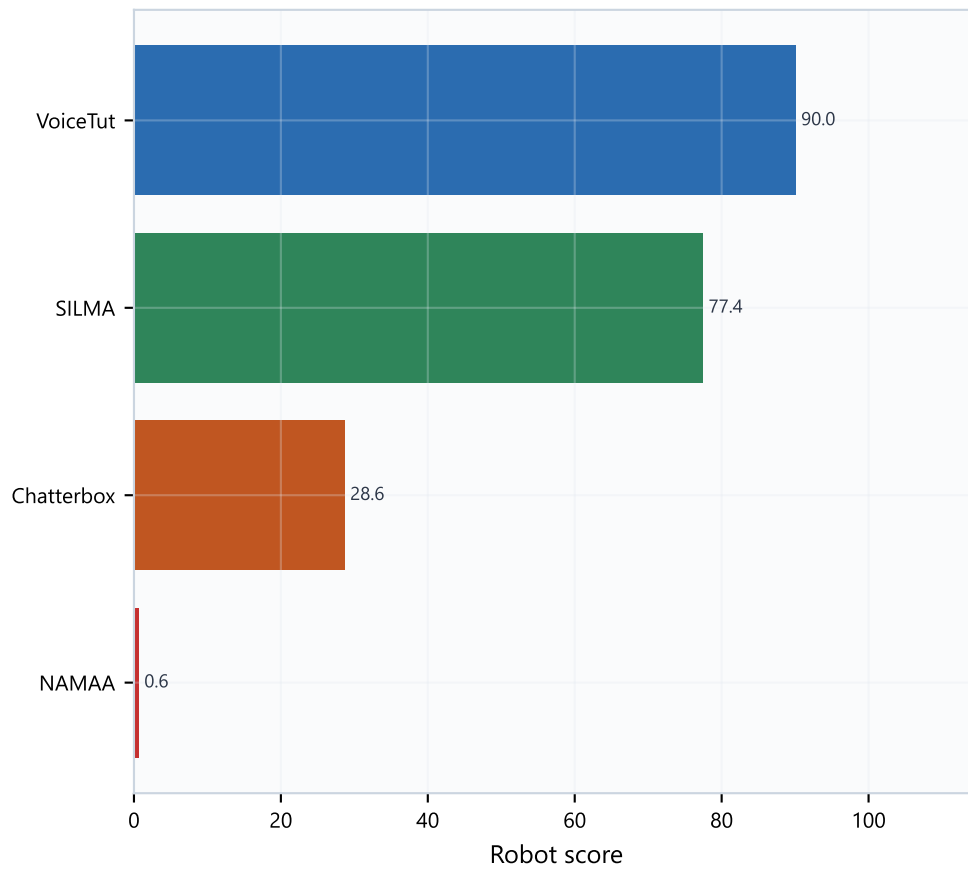
Accuracy alternative: Whisper-Large-v3-CT2.

Whisper CT2 family occupies the useful top band; Qwen ASR variants lag badly on WER.

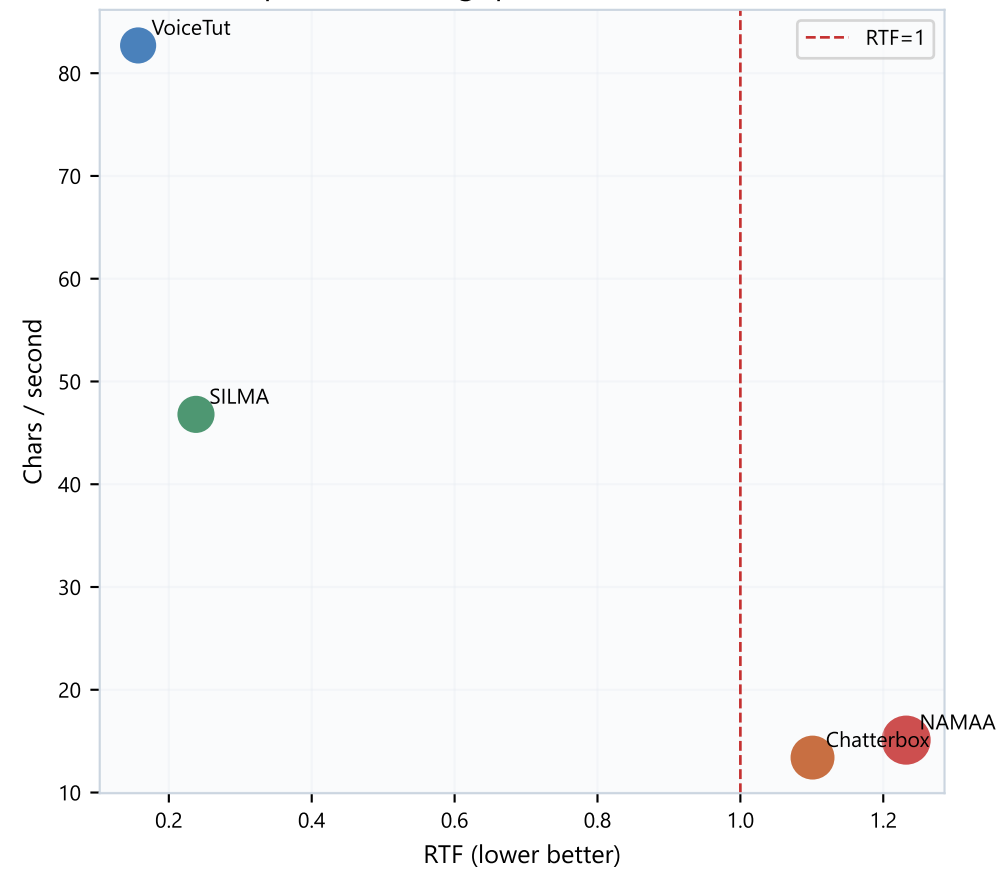
Keep ASR VRAM low so the LLM + TTS still fit on the same GPU.

TTS snapshot

TTS robot leaderboard



Speed vs throughput (bubble = VRAM)



TTS takeaway

Default speed/resources pick: VoiceTut-TTS.
SILMA-TTS is the best realtime alternative if VoiceTut fails the listen test.
Chatterbox and NAMAA are not realtime on this bake-off — use only if voice quality clearly wins.
Always listen to WAVs before freezing the production voice.

Package contents & bottom line

PDF package

LLM_Model_Selection_Report.pdf — methodology + charts + quality guidance
ASR_Model_Selection_Report.pdf — methodology + accuracy/speed evidence
TTS_Model_Selection_Report.pdf — methodology + picks + listening checklist
Combined_Families_Summary.pdf — this cross-family overview + VRAM stack budget

WAV files for listening: analytics/final/tts_outputs/*.wav

Bottom line

ASR = Whisper-Large-v3-Turbo-CT2 (or Whisper-Large-v3-CT2 for max accuracy).
LLM = Nile-Chat-4B (speed) or Qwen3-8B (quality).
TTS = VoiceTut-TTS pending listening confirmation (SILMA as backup).
VRAM plan: speed stack ~13.7 GB · quality stack ~12.2 GB (T4 ~14.9 GB).
See family PDFs for full evidence before final sign-off.